

# Test 12: Real zeros vs GUE matrix

Verdict: FAIL | Generated: 2026-09-14 23:56:59 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

## What this test verifies

Two-sample KS: data vs simulated GUE matrix spectrum.

METHOD: Two-sample KS: data vs simulated GUE matrix spectrum.

## Result

Test 12 [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

plots/plot\_01.png ? 1600×1000, ABPlotV23 engine (supersampled)

## Plots

- plots/animation.gif ? animated reveal of every plot of this test

Reproduce with: julia ab\_cloud\_v23.jl --test 12 --no-two-pass

## Independent verification guide

Every computation is logged in logs/computation\_log.txt (timestamps + deltas); raw console output in logs/stdout\_capture.txt.

(no log\_comp entries)

## FULL COMPUTATION LOG

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## CONSOLE CAPTURE

TEST 12 (secondary, HARDCORE): Objection 2 (extended): Advanced RMT Diagnostics ? 2KS symmetry audit

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Full: real n=8400 vs ref n=8400, D=0.026548, p=0.005252 (primary verdict governs)

W1 ref split : D\_vs\_A=0.024405, D\_vs\_B=0.030238, |?D|=0.005833 (limit 0.02)

W2 blocks : 0/4 blocks agree with full verdict at ?=0.01 (limit ? 3)

W3 effect : D\_full=0.026548 < 2KS critical at ?=0.01: 0.025151? NO

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W1 ref symmetry PASS ? |?D| = 0.005833

W2 block agree WARN ? 0/4 agree

W3 D bound WARN ? D 0.0265 vs crit 0.0252

Test 12 [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN